Geotagged Airline Tweets

Completed activity report | dataset and query results

A. Dataset exploration

The full Twitter US Airline Sentiment CSV has 14,640 rows. Of these, 1,019 contain a non-empty *tweet_coord*; 164 are the placeholder [0.0, 0.0]. That leaves 855 valid coordinate rows before indexing.

B. Indexing results

The indexer processes every CSV row, but *tweet_id* is the Elasticsearch document ID. The source contains 155 repeated IDs, so later rows update an existing document. This is why the successful bulk action count is 14,640 while the final index contains 14,485 unique documents. There are 847 unique indexed documents with valid locations after the duplicate-ID overwrites.

Check	Result
Full CSV rows read	14,640
Rows skipped	0
Bulk actions indexed / failed	14,640 / 0 expected successful actions
CSV rows with valid coordinates	855
GET /tweets/_count (unique IDs)	14,485

B2. Bulk API sends many documents in fewer requests, reducing network overhead. Using *tweet_id* as *_id* makes reruns update the same document and prevents duplicates. [0.0, 0.0] is a placeholder, not a real GPS position, so removing it avoids false geographic matches.

C. Full-text and exact-value queries

Query	Calculated result
Q1	9,088 negative tweets; 2,633 negative United tweets.
Q2	1,925 tweets match the query terms. The five most recent matches are listed below.
Q3	90 positive tweets contain the exact phrase "customer service".
Q4	25 negative Delta tweets have reason Late Flight on Feb 23, 2015 (Pacific time).
Q5	Top retweet counts: 44, 32, 31, 28, 22.

Q2 examples (tweet ID | airline | sentiment): 570307026263384064 | United | negative; 570303797605617665 | Southwest | negative; 570301395141836801 | American | negative; 570298018685165568 | United | negative; 570291492545298432 | United | neutral.

Q4 examples: 570066554848743424, 570066245195829249, 570056726793359360, 570048890466111488, 570039630973829120. **Q5 top tweets (ID | retweets | airline):** 567897883875217408 | 44 | US Airways; 567909106553483264 | 32 | US Airways; 569927288751587328 | 31 | Delta; 569584057270575105 | 28 | US Airways; 568869594531651584 | 22 | Southwest.

Q2 text excerpts: United: delayed again because of deicing; Southwest: flight delayed four hours; American: 304-minute delay and missed connection; United: lost car seat; United: lost voucher.

Q5 returned text (retweets | airline | excerpt): 44 | US Airways | 5-hour delay, wants to go home; 32 | US Airways | never again, will choose wiser; 31 | Delta | stop using fleek; 28 | US Airways | throwback livery; 22 | Southwest | beautiful day in Seattle.

C. Aggregations

Q6 Airline	Total	Negative	Neutral	Positive
American	2,604	1,870	429	305
Delta	2,222	955	723	544
Southwest	2,420	1,186	664	570
US Airways	2,913	2,263	381	269
United	3,822	2,633	697	492
Virgin America	504	181	171	152

Q7 Top negative reasons per airline:

Airline	Top 3 reasons (count)
American	Customer Service Issue (741); Late Flight (237); Cancelled Flight (231)
Delta	Late Flight (269); Customer Service Issue (199); Can't Tell (186)
Southwest	Customer Service Issue (391); Cancelled Flight (162); Can't Tell (159)
US Airways	Customer Service Issue (811); Late Flight (453); Can't Tell (246)
United	Customer Service Issue (681); Late Flight (525); Can't Tell (379)
Virgin America	Customer Service Issue (60); Flight Booking Problems (28); Can't Tell (22)

Q8 Daily counts (Pacific): Feb 16: 4; Feb 17: 1,408; Feb 18: 1,344; Feb 19: 1,376; Feb 20: 1,500; Feb 21: 1,557; Feb 22: 3,073; Feb 23: 3,026; Feb 24: 1,197. Busiest day: Feb 22 (3,073).

Q9 Top hashtags: #destinationdragons (81), #fail (67), #jetblue (48), #unitedairlines (45), #customerservice (36), #usairways (28), #neveragain (27), #americanairlines (26), #united (26), #usairwaysfail (26).

D. Geographic queries

Query	Calculated result
Q10	847 of 14,485 indexed documents have a valid location (5.85%).
Q11	83 tweets are within 25 km of JFK. Nearest five: 0.293 km (568611474416906240); 0.423 km (569967019958730753); 0.471 km (568216703055876096); 0.471 km (568204705861652480); 0.472 km (568196466503770112).
Q12	80 negative geotagged tweets are inside the California box. United leads with 43, followed by American 13, Southwest 13, US Airways 6, Delta 4, Virgin America 1.
Q13	Around O'Hare: 0-50 km = 47; 50-500 km = 24; over 500 km = 776.
Q14	Top precision-6 cells: 6/18/24 (205), 6/16/23 (53), 6/17/25 (53), 6/11/25 (46), 6/15/26 (46). Negative geotweet centroid: 39.1647, -89.0527 (567 points).

Method note: the calculations use the full CSV and the indexer's cleaning rules, including last-write-wins behavior for duplicate tweet IDs, Pacific-time bucketing, the specified geo boundaries, and unique lowercase hashtags. The sample dry-run was executed and matched the activity expectation: 5 rows read, 0 skipped, 2 valid coordinates; the [0.0, 0.0] row has no location. This workspace had no connected Elasticsearch cluster, so the full-run counters and query results are computed against the dataset and should be confirmed by running queries.txt on the student cluster. Dataset source: Twitter US Airline Sentiment (CrowdFlower), CC BY-NC-SA 4.0.